

available at www.sciencedirect.comjournal homepage: www.elsevier.com/locate/biocon

Physiological and reproductive consequences of human disturbance in Humboldt penguins: The need for species-specific visitor management

Ursula Ellenberg^{a,c,*}, Thomas Mattern^c, Philip J. Seddon^c, Guillermo Luna Jorquera^{a,b}

^aDepartamento de Biología Marina, Facultad de Ciencias del Mar, Universidad Católica del Norte, Larrondo 1281, Coquimbo, Chile

^bCentro de Estudios Avanzados en Zonas Áridas CEAZA, Larrondo 1281, Coquimbo, Chile

^cZoology Department, University of Otago, 340 Great King Street, Dunedin, New Zealand

ARTICLE INFO

Article history:

Received 14 November 2005

Received in revised form

9 May 2006

Accepted 17 May 2006

Available online 30 June 2006

Keywords:

Ecotourism

Human disturbance

Heart rate telemetry

Habituation

Spheniscus humboldti

Chile

ABSTRACT

Over the last decade the Humboldt penguin, *Spheniscus humboldti*, has become a focus for ecotourism. Current management applies visitor guidelines similar to those developed for Magellanic penguins, *Spheniscus magellanicus*. However, unlike these, Humboldt penguins are extremely sensitive to human presence. Breeding success was significantly reduced at frequently visited sites. Heart rate telemetry during disturbance experiments revealed that Humboldt penguins respond more strongly to human presence than do any other penguin species thus far studied. A person passing an incubating penguin at 150 m distance already provoked a significant heart rate response. Recovery times were up to half an hour after direct human approach, causing increased energy expenditure without any overt behavioural reaction. Being extraordinary sensitive to human activity with little habituation potential the Humboldt penguin proves to be a difficult focal species for ecotourism. For sustainable ecotourism visitors are required to stay out of sight of Humboldt penguin breeding and moulting areas. Management guidelines need to acknowledge that even closely related species may react very differently towards human presence.

© 2006 Elsevier Ltd. All rights reserved.

RESUMEN

En el transcurso de la última década el pingüino de Humboldt (*Spheniscus humboldti*) se ha transformado en un foco de atención para el ecoturismo. El manejo actual que se aplica a los visitantes es similar a aquella desarrollada para el pingüino de Magallanes (*Spheniscus magellanicus*). Sin embargo, a diferencia de éste último, el pingüino de Humboldt es extremadamente sensible a la presencia humana. El éxito reproductivo se redujo significativamente en los sitios más visitados. Mediante métodos de telemetría medimos la frecuencia cardíaca frente a disturbios experimentales y determinamos que los pingüinos de Humboldt reaccionan a la presencia humana en forma más intensa que las otras especies de pingüinos estudiadas hasta ahora. En pingüinos incubando una persona caminando a 150 m de distancia del nido, provocó un aumento significativo de la frecuencia cardíaca. El tiempo de recuperación frente a una persona aproximándose directamente,

* Corresponding author. Present address: Zoology Department, University of Otago, 340 Great King Street, P.O. Box 56, Dunedin, New Zealand.

E-mail addresses: ulnberg@eudypes.net (U. Ellenberg), t.mattern@eudypes.net (T. Mattern), philip.seddon@stonebow.otago.ac.nz (P.J. Seddon), gluna@ucn.cl (G.L. Jorquera).

0006-3207/\$ - see front matter © 2006 Elsevier Ltd. All rights reserved.

doi:10.1016/j.biocon.2006.05.019

fue de más de una media hora provocando un incremento del gasto energético sin que el animal mostrará ninguna reacción conductual evidente. Debido a su sensibilidad extrema y bajo potencial de habituación, el pingüino de Humboldt ha probado ser una especie focal difícil para el ecoturismo. Para un manejo sostenido de ésta actividad en periodos reproductivos y en áreas de muda, los visitantes deben mantenerse fuera de la vista de los pingüinos. En la elaboración de reglas de manejo se debe admitir que aún tratándose de especies altamente relacionadas, éstas pueden reaccionar de manera muy diferente frente a la presencia humana.

© 2006 Elsevier Ltd. All rights reserved.

1. Introduction

Conservationists place great hopes on ecotourism “as responsible travel to natural areas which conserves the environment” (Lindberg and Hawkins, 1993); “producing economic benefits that encourage conservation” (Ryel and Grasse, 1998). There is ample evidence, however, that so-called ecotourism has the potential to be ecologically unsustainable (e.g. Edington and Edington, 1986; Boo, 1990; Honey, 1999; Müllner et al., 2004).

Although adverse effects of human disturbance on wildlife have long been recognized (e.g. Liddle and Scorgie, 1980; Boyle and Samson, 1985; Ingold et al., 1992; Keller, 1995; Carney and Sydeman, 1999), a general framework for approaching this issue is lacking (Frid and Dill, 2002). Effects on survival or breed-

ing success are often seen as the ultimate criteria for identifying adverse effects (Nimon and Stonehouse, 1995; Nisbet, 2000). However, even subtle impacts of human disturbance on physiological parameters, such as increased heart rate, stress hormone levels and energy expenditure, may reduce individual fitness and can eventually have population-level consequences (Lee and McDonald, 1985; Hüppop, 1995; Regel and Pütz, 1997; Fowler, 1999; Frid and Dill, 2002; McClung, 2004).

Penguins may show little behavioural reaction to human presence at their breeding sites (Culik and Wilson, 1991; Nimon et al., 1995). Heart rate change can be a reliable parameter to quantify penguin stress response to single disturbance events (Wilson et al., 1991; Nimon et al., 1996; Giese et al., 1999). Although physiological stress responses in penguins

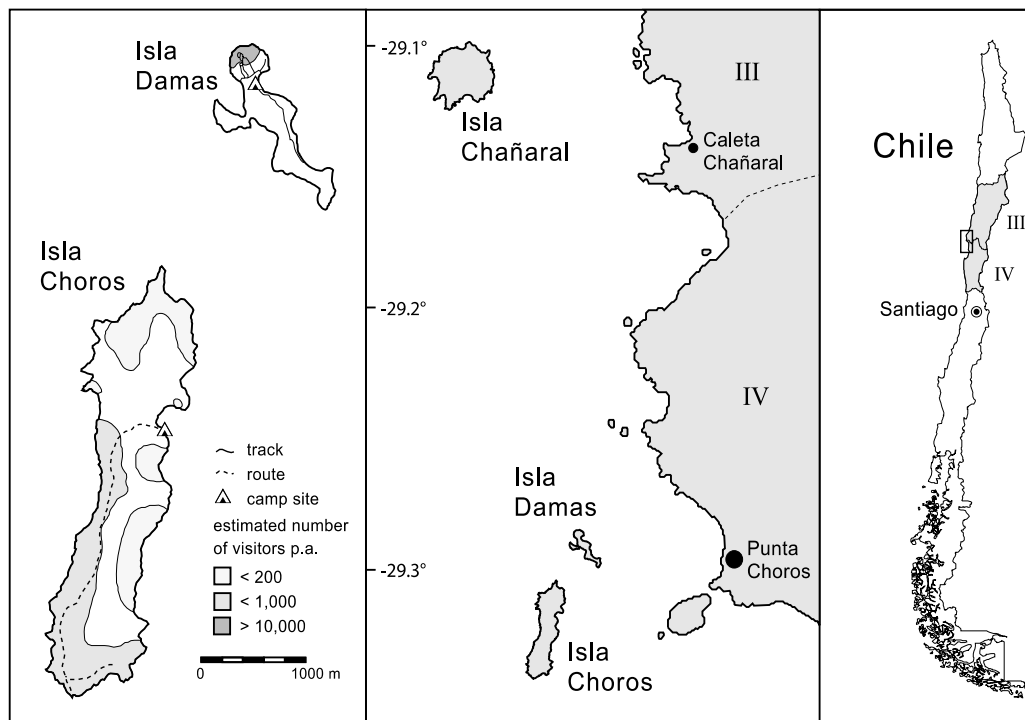


Fig. 1 – Geographic overview of Isla Choros, Damas and Chañaral, three islands which form the Humboldt Penguin National Reserve in north-central Chile. The administrative regions Coquimbo (4th region) and Atacama (3rd region) are shaded grey and labelled with roman numbers. Tourism mainly operates from Punta Choros where the reserve's head quarters (CONAF – Corporación Nacional Forestal) are located. Caleta Chañaral is a small fishing village facing Isla Chañaral. The close up on Isla Choros and Damas reveals main Humboldt penguin breeding areas and associated levels of visitation as well as access facilities.

have not been linked to fitness parameters, there is evidence human disturbance may cause reduced breeding success in several penguin species, including African *Spheniscus demersus* (Hockey and Hallinan, 1981), Yellow-eyed *Megadyptes antipodes* (Roberts and Roberts, 1973), Gentoo *Pygoscelis papua* (Croxall et al., 1984) and Adélie penguins *P. adeliae* (Woehler et al., 1994; Giese, 1996).

Humboldt penguins *Spheniscus humboldti* have become a focus for tourism ventures relatively recently. This has been in part due to the remoteness of colonies which are mainly restricted to islands and inaccessible sea caves along the coasts of Chile and Peru (43° S to 5° S). The Humboldt penguin is classified as vulnerable (IUCN, 2004) and although legally protected is still hunted for food and bait (BirdLife International, 2005).

The Humboldt Penguin National Reserve (29° S), IV Region, Chile, including the islands Damas, Choros and Chañaral (Fig. 1) is the premier breeding location of the Humboldt penguin (Simeone et al., 2003; Mattern et al., 2004). Since the reserve's creation in 1990 associated tourism has developed rapidly, but with few controls. Current code of practice applies management guidelines similar to those designed for visitation of Magellanic penguin *Spheniscus magellanicus* colonies, a species which appears to be relatively tolerant of people in close proximity to nests (Yorio and Boersma, 1992). This, however, assumes that different species will react similarly towards human presence. Current lack of knowledge about the species' sensitivity to human proximity leaves the Humboldt penguin vulnerable to mismanagement (Vilina et al., 1995; Simeone and Schlatter, 1998; Simeone et al., 2003).

In this study, we combined two approaches to assess human disturbance effects on penguins in the Humboldt Penguin National Reserve. Using heart rate telemetry we experimentally quantified the magnitude and duration of physiological responses to disturbance stimuli and evaluated the penguins' potential for habituation. Additionally, we monitored breeding performance of Humboldt Penguins exposed to different levels of human activity, considering other factors which might affect reproductive output. We sought to provide a robust species-specific basis for effective management of human activity in the proximity of Humboldt penguins.

2. Methods

2.1. Defining “human disturbance” and “stress”

The term “human disturbance” is generally associated with human activity that has negative consequences for wildlife. As “negative” is subjective and difficult to quantify, we adopted the definition by Nisbet (2000): “Human disturbance is any human activity that changes the contemporaneous behaviour or physiology of one or more individuals [...]”. Hence, disturbance causes by definition an effect, however, whether this effect is sufficiently adverse to require management action remains to be determined.

“Stress” is used in accordance with following definition provided by McEwan (2000) “Stress may be defined as a real or interpreted threat to the physiological or psychological integrity of an individual that results in physiological and/or

behavioural [stress] responses.” For a detailed discussion of contemporary stress definitions refer to Levine (2005). To avoid confusion we distinguish between the input (stress or disturbance stimulus) and the output (stress response).

2.2. Study site and species

During the austral summer breeding seasons 2001/02 and 2002/03 we studied Humboldt penguins on the islands Damas, Choros, and Chañaral (29° S, Chile, Fig. 1), which form the Humboldt Penguin National Reserve. While public access to Choros and Chañaral is restricted, Damas has become an important tourism destination, and visitor numbers have grown from a few hundred people annually in the early 1990s to over 10,000 visitors in 2003 (CONAF, Coperación Nacional Forestal, responsible for administration of the islands, unpublished data). Being more remote, Chañaral is still rarely visited by humans. Choros and Damas, on the other hand, are situated within easy reach from the popular tourist village Punta Choros where local fishermen and tourism companies offer wildlife-viewing trips. During the main tourist season (November–March) more than 40 tours depart from Punta Choros on most days, usually including a stop-over of several hours on Damas to explore local wildlife as well as private beaches with their diverse desert flora. CONAF tries to limit landings on Damas to 100 tourists per day. Damas also features overnight camping facilities, and since 2002, visitors are asked to stay no longer than two nights on the island due to increasing demand for camp sites. Damas' penguin breeding area has access via a well established loop track (Fig. 1). Similarly as in Magellanic penguin colonies used for tourism ventures, visitors can walk freely among nests and approach penguins to within few meters of nest sites. Although landing on Choros is officially not permitted, fishermen frequently use the island as base for seafood harvest or hunt for rabbits. In recent years, the island received increasing scientific attention with research camps being established throughout the summer months, addressing a wide variety of scientific questions. However, our group was the first to work with the local Humboldt penguin population.

Humboldt penguins breed in loose colonies and are resident on the islands all year. Most nests are in dug-out bowls under rocks or vegetation. Humboldt penguins lay two eggs per clutch, but three-egg clutches occur naturally (1.2% of 262 nests, own observation). Chicks hatch after 41 days and are reared by both parents until fledging at 10–12 weeks (Zavalaga and Paredes, 1997). During moult penguins are confined to land for three weeks (Luna Jorquera et al., 2000). The peak breeding (November–December) and moulting (February) take place during summer when the reserve is a focus for wildlife tourism.

2.3. Breeding parameters and nest attendance pattern

We monitored nests on Choros and Damas during 2001/02 and 2002/03, and on Chañaral in 2002/03. Established human walkways were geo-referenced. We determined GPS positions (± 10 m accuracy) of nests, and recorded nest type (rock boulders, rock cave, cactus, and shrub), entrance orientation (azi-

mouth from North), height above sea level (metres), and estimated the percentage cover of the nest bowl to investigate the influence of various nest characteristics on breeding performance. During nest searches we also recorded roosting sites (containing adults only), and empty nests (with signs of previous usage like scat marks or feathers). In active nests we noted presence of external parasites (ticks, fleas, and mites). Ticks were quantified by sieving through 500 ml of nest bottom substrate and/or placing a cup with 100 ml butyric acid for 24 h into the nest bowl after confirmed nest abandonment. We measured egg length and maximum width with vernier callipers (± 0.1 mm). Egg size was estimated using a volume index ($\text{length} \times \text{maximum width}^2$) to test for differences in female condition and/or food availability between breeding sites (Boersma et al., 1990). Chicks were weighed with Pesola balances (1, 2 and 5 kg). Following Boersma et al. (1990) we considered a chick to have fledged when it was no longer found in the vicinity of its nest site and weighed ≥ 1.8 kg when last seen. At a subset of nests ($n = 13$) that were easily observable from distance we recorded adult nest attendance patterns. For this, we colour-marked one attending bird with a long-handled paint brush. As Humboldt penguins demonstrate an obvious difference between sexes in that males have bigger bills and are significantly larger than females (Williams, 1995), birds were sexed via visual comparison of joint partners. Over a period of two weeks the gender of the attending adult was noted daily in the early afternoon. Humboldt penguins usually depart colony after sunrise and return from their foraging trip in the evening (Williams, 1995) so that error margins of attendance patterns should be negligible. To reduce researcher disturbance we checked nest sites once per month from October until February.

2.4. Heart rate telemetry

Heart rate telemetry was carried out on Choros at a total of 20 nest sites (10 per breeding season, with five nests in more frequented and five in less frequented parts of the island, Fig. 1). Only birds which did not readily flee and abandon their nest site during human approach were included in disturbance experiments. All experimental nests contained two egg clutches. To record heart rate (HR) we used an egg-shaped dummy (ED) with internal microphone (Panasonic MCE-201) and pre-amplifier (Conrad Electronic 191566-62). Via 100 m acoustic cable (0.75 mm^2) the HR signal was recorded to MiniDisc (Sharp MD-MT88). Simultaneously, we noted behavioural changes, natural stimuli (e.g. approaching partner or neighbour) and disturbance experiment details.

EDs were added to the clutch when only one partner was present. During deployment we either colour-marked the incubating bird with a long-handled paint brush or noted individual spot patterns to distinguish the bird which was present during ED deployment from its naïve partner. From four available EDs a maximum of two were deployed per day, retrieval depended on the time of return of the naïve partner. EDs were disinfected between deployments. After ED deployment we observed the bird from a hide (50–100 m from nest amongst rock boulders) but did not start experiments until the next day.

The ED was accepted by 94% of males ($n = 18$) and 82% of females ($n = 17$). One male fled during approach and did not return until the next day; one female rejected the ED after one day of incubation; two females never accepted the ED when relieving their partner. When accepted, the ED was incubated within the clutch; in no case were real eggs expelled.

Sound data was analyzed with custom written software in Matlab 6.5 (Mathworks, Natick, MA, USA). As resting heart rate (RHR) was found to be as low as 50 bpm (beats per minute), we considered a window of 12 s (containing a minimum of 10 heart beats) to calculate each heart rate reading. Following Neebe and Hüppop (1994) we defined the mean RHR during a period of at least 30 s of undisturbed incubation immediately before the experiment or natural stimulus as baseline; two standard deviations from mean RHR were considered a tolerance band. When the HR left the tolerance band spontaneously it counted as excitation. Excitation ended when the HR was maintained for at least 30 s within the previously defined tolerance band. Recovery time was defined as the time from maximum HR to end of excitation.

We recorded HR during experimental disturbance, involving a person walking past the nest at 150, 100, 50, and 20 m distance within sight of the incubating bird, and direct approaches to 20 or 2 m from the nest. During three trials, birds responded equivalently when directly approached to both distances (%RHR: Mann Whitney $U = 13$, $p = 0.529$; recovery time: $U = 17$, $p = 1$). Therefore, we abandoned the 20 m direct approach to reduce disturbance impact and subsequently removed this experiment from comparative analysis. Instead, we focused on 2 m direct approaches, which reflect a typical nest-check situation when no interference is required. This nest-check experiment included a one minute stop in front of the nest before retreating out of sight. During the stop we counted the number of alternate stares, a typical behaviour of defensive *Spheniscus* penguins, which was first described as “alternate head stare” by Eggleton and Siegfried (1979) and subsequently used to quantify behavioural stress response (e.g. Yorio and Boersma, 1992; Fowler, 1999; Walker et al., 2006). We also included a pass at 20 m distance, but out of sight of the incubating bird to determine whether the sound of a walking person elicits a response. Experiments were performed by a single person dressed in inconspicuous clothing moving in a calm steady manner (i.e. slow and even steps < 1 /s; no quick movements); all direct approaches were done by UE. Individual penguins were exposed to several passes but only one direct approach per day.

To assess whether penguins readily habituate to repeated human presence we recorded HR during three consecutive passes at 20 m distance, allowing 30 min between experiments to ensure HR had returned to resting levels before starting the next pass. Additionally, we measured the differences between the same experiment performed on subsequent days, and assessed whether physiological responses decreased over time.

2.5. Statistical analysis

For multiple comparisons, analysis of variance (ANOVA) was employed. Following significant ANOVA results, we used Tu-

key–Kramer honestly significant difference as our criterion for significance (Zar, 1999). We square-root-transformed recovery time data when assumptions of normality were not met and tested for homogeneity of variances using Levene's test. Kruskal–Wallis test was employed to confirm non-significant ANOVA results when sample sizes were different. Repeated-measures ANOVA or paired t-test were employed for analysis of habituation experiments. A pooled variances two-sample t-test for two-tailed hypothesis was applied for comparison of means, if not indicated otherwise. Pearson correlation coefficient was calculated to quantify correlation. We considered differences significant if $p < 0.05$ and reported values as mean (range), if not indicated otherwise.

3. Results

3.1. Nest attendance and ED deployment time

Nest attendance patterns were extremely variable: Considering successful nests only, mean male incubation shifts were 2.5 days ($n = 16$, range 1–8 d), and those of females 3.1 days ($n = 15$, range 1–6 d). As we were interested not only in the bird which had been present when we deployed the egg-shaped dummy (ED) but also its naïve partner, we extended deployment time to an average of 4 days (range 2–10 d). There was a trend for lower breeding success with increasing ED deployment time ($n = 19$, $r = -0.28$, $p = 0.25$).

3.2. Breeding parameters

Along the south-western plateau of Choros, an area frequented by humans (see Fig. 1) we found a low number of active nests despite abundant nesting opportunities. Of 710 nest sites only 6% were occupied by breeders. In comparison, the northern breeding area on Choros held a total of 221 nest sites with 20% of them active. This is comparable to the occupation rate found on undisturbed Chañaral (14%).

Only one fourth of nests on Choros had less than 50% overhead cover. Breeding success of less sheltered nests did not

differ significantly from more protected ones, although the only two nests without any cover failed.

Choros provided diverse nesting habitat with most nests being established under rock boulders (74%), followed by rock caves (10%), shrub (9%) and cactus (6%). We found no significant differences in breeding success between different nest types ($F_{3,132} = 1.931$, $p = 0.128$) with a trend for cactus being a suboptimal nesting habitat. Chañaral offered little choice of nesting habitat; cactus was used as nesting substrate in more than half of all cases (52%), followed by shrub (48%). On Damas all nests were found under rock boulders.

Height above sea level or direction of main nest entrance did not correlate with breeding success on any of the islands. External parasite infection of nests was low at all sites.

Mean ($\pm$ SD) egg size index calculated from measurements was 215.4 ± 17.6 with no difference between the three islands ($H_{2,178} = 1.766$, $p = 0.413$).

Birds attending relatively open nests occasionally fled during nest-checks but usually returned soon. In one incident the bird completely abandoned its two egg clutch during a nest-check, thus one single visitation of a nest can already cause breeding failure in Humboldt penguins.

Damas had the lowest breeding success during both seasons monitored; all nests visible from human walkways failed. The breeding success on Choros did not differ significantly between the relatively undisturbed and the more frequently visited areas during both breeding seasons (2001/02: $t_{84} = -0.815$, $p = 0.417$; 2002/03: $t_{110} = -1.159$, $p = 0.249$), thus breeding data collected on Choros was pooled despite different disturbance regimes. On Choros penguins had better reproductive output than on Damas but less than on remote Chañaral, which had the highest breeding success of all areas (1.34 chicks fledged per pair, Table 1). In 2002/03 breeding success was significantly different between all three islands ($F_{2,221} = 11.292$, $p < 0.001$; Choros vs. Chañaral $p = 0.017$, Choros vs. Damas $p = 0.002$, Damas vs. Chañaral $p < 0.001$). Breeding success of experimental nests ($n = 10$ per season) was 1.1 chicks per nest in 2001/02 and 0.7 in 2002/03 breeding season.

Table 1 – Breeding success (chicks fledged per pair) and number of active nests reflect levels of human activity on the three islands (Damas, Choros and Chañaral) forming the Humboldt Penguin National Reserve

Island	Damas	Choros ^a	Chañaral
Surface area (ha)	56	322	655
Estimated number of visitors (pa)	>10,000	<1000	≤100
Estimated number of active nests	10 ^b	360 ^b	9000 ^c
2001/02 breeding success	0.44	0.81	
Number of nests fate known	9	93	
2002/03 breeding success	0.13	0.96	1.34
Number of nests fate known	8	122	82

Numbers of active nests include only nests established during the austral summer breeding peak (October–December).

During 2001/02 differences in breeding success observed on Damas and Choros were almost significant (separate variances $t_{12} = 1.859$, $p = 0.088$). During 2002/03 breeding success was significantly different between all three islands ($F_{2,212} = 14.683$, $p < 0.001$; posthoc, Choros vs. Chañaral $p = 0.003$, Choros vs. Damas $p = 0.014$, Damas vs. Chañaral $p < 0.001$).

a Excluding experimental nests.

b See Simeone et al. (2003).

c See Mattern et al. (2004).

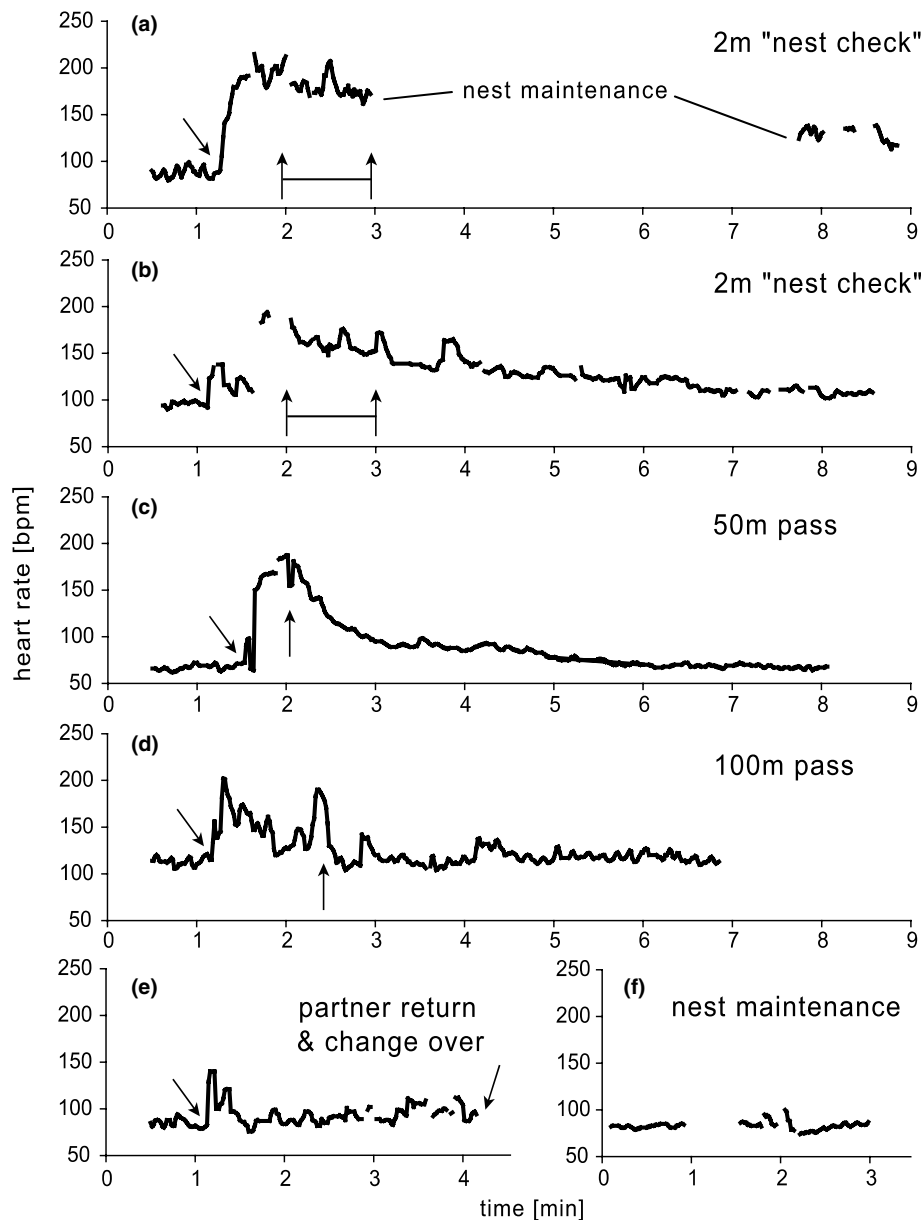


Fig. 2 – Examples of heart rate change over time during experimental (a–d) and natural (e and f) disturbance stimuli. The first arrow indicates point of first visual contact. The nest check experiment (a direct approach to within 2 m from the incubating bird) included a 1 min stop (two joined arrows indicate the time spent motionless at the nest). During experimental passes at 50 or 100 m distance the second arrow marks the point of minimum distance from the nest site (c and d); while during partner return (e) it marks the change over. Gaps in curves are a result of temporary loss of heart rate signal due to penguin movements or high ambient noise levels (e.g. partner recognition calls).

3.3. Heart rate telemetry

During experiments we recorded an average resting heart rate (RHR) of 104 beats per minute (bpm, $n = 59$, range: 65–142 bpm). RHR was independent of sex, date or time of day, but varied with nest microclimate. Therefore we used the relative HR increase (%RHR) to determine levels of excitation. HR response and recovery times during nest-check experiments were comparable during both seasons (%RHR: $t_{15} = 0.623$, $p = 0.453$; recovery time: $t_{15} = 1.644$, $p = 0.121$). Similarly, we

found no significant differences in heart rate responses of birds breeding in the rarely visited and the more frequented part of Choros (%RHR: $t_{15} = 0.796$, $p = 0.438$; recovery time: $t_{15} = 1.385$, $p = 0.186$). Thus, HR response and recovery data from both areas were pooled for subsequent analysis.

3.3.1. Disturbance experiments

A direct approach to within 2 m of the nest site (i.e. nest-check experiment) provoked the strongest physiological reaction (Figs. 2 and 3): The birds doubled their HR (198%RHR,

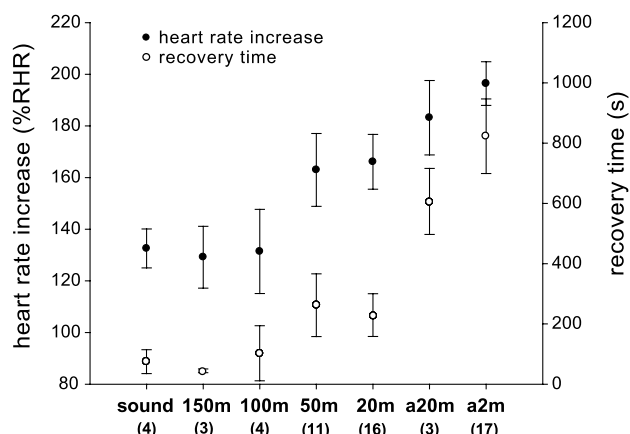


Fig. 3 – Differences in heart rate elevation (in percent of resting heart rate, %RHR) and time needed for recovery after different experimental disturbance stimuli including sound (a person passing the nest at 20 m distance without visual contact), passes at 150, 100, 50 and 20 m distance from the nest site, and direct approaches to within 20 and 2 m (i.e. nest check experiment) from the incubating bird. Error bars represent standard error of the mean, for each experiment sample size is given in brackets.

range 152–287, $n = 17$) and needed up to half an hour for recovery (13.7 min, range 3.7–29.9 min, $n = 17$). This response was significantly different from reactions recorded during all other experiments (%RHR: $F_{3,50} = 7.232$, $p < 0.0001$; recovery

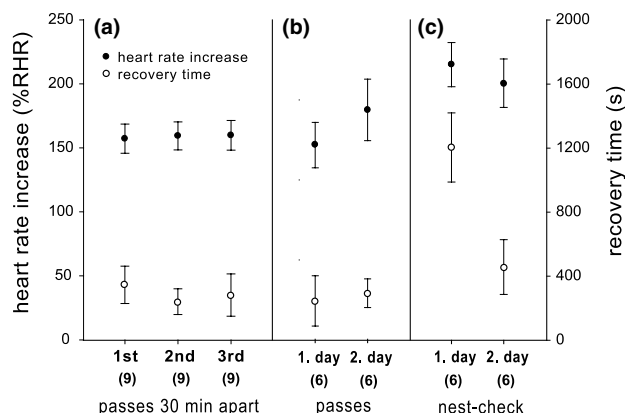


Fig. 4 – Humboldt Penguins exhibit a low potential for habituation to human disturbance. (a) Three consecutive passes in 20 m distance (30 min apart) provoked similar stress responses during each pass, as well as (b) first passes on consecutive days resulted in comparable stress responses. (c) After a direct approach to 2 m from the nest (i.e. nest-check experiment) during two consecutive days recovery time was significantly reduced the second day, although no difference in heart rate reaction was found. Heart rate elevation is given in percent of resting heart rate, %RHR, recovery time is the time needed until heart rate returned to pre-disturbance levels. Error bars represent standard error of the mean, for each experiment sample size is given in brackets.

time: $F_{3,50} = 10.528$, $p < 0.0001$). For analysis we combined 100 m and 150 m experimental passes into ‘ ≥ 100 m pass’ and excluded the three 20 m direct approaches (see Section 2). Direct approaches resulted in a stronger HR response than did movements clearly leading past the nest (Fig. 3). Nevertheless, even a person passing at 150 m distance caused a measurable response (Fig. 3). An approaching partner caused the strongest recorded natural HR reaction (164%RHR, SD 4.7, $n = 3$). During natural stimuli, HR usually fell back into range of RHR within seconds (e.g. Fig. 2e and f), whereas human-induced responses needed more time for recovery (Fig. 2a–d).

During nest-check experiments 71.4% ($n = 20$) of the birds stood up and thus lost contact with the ED at some stage. This is twice as often as during all other experiments (14.3–33.3%). Average stand-up distance was 20.2 m (range 7–50 m) during nest-check experiments; including passes it increased to 35.5 m (range 7–110 m). Loss of contact with the ED immediately after an experiment was usually followed by vigorous nest maintenance behaviour (e.g. Fig. 2a) however even birds which did not display a strong behavioural reaction had comparable HR responses to a nest-check experiment (e.g. Fig. 2b).

We counted an average of 29 alternate stares during the 1 min nest-check (range 16–53, $n = 19$). Even though HR did not fall significantly during the check, head turns were on average quicker (17 turns) during the first 0.5 min than during the second half of the stop (11 turns; separate variances $t_{26.8} = 2.924$, $p = 0.007$). Overall speed of head turns was slightly positive related with heart rate response (%RHR: $r = 0.308$, $p = 0.246$, $n = 16$; recovery time: $r = 0.386$, $p = 0.139$, $n = 16$). In penguins breeding along the south-western plateau head turns were on average slower (although not significantly so: separate variances $t_{17} = 1.746$, $p = 0.106$) and less variable (coefficient of variation $CV_{\text{north}} = 0.44$, $CV_{\text{south}} = 0.14$) when compared with the undisturbed northern colony.

We found no significant difference between sexes when comparing maximum HR, %RHR, recovery time, head turns and likelihood to stand up during a nest-check experiment (data available for 8 females and 12 males). During all experiments responses of naïve birds ($n = 10$) were not different from that of mates which experienced ED deployment ($n = 10$).

Breeding success of experimental nests was unrelated to individual heart rate response, considering reactions to 2 m direct-approach and only the shyer partner per nest (%RHR: $F_{2,15} = 0.430$, $p = 0.660$; recovery time: $F_{2,15} = 0.420$, $p = 0.666$).

3.3.2. Habituation potential

Three consecutive passes at 20 m distance (30 min apart) provoked similar stress responses during each pass (%RHR: $F_{2,24} = 0.056$, $p = 0.946$; recovery time: $F_{2,21} = 2.241$, $p = 0.202$; Fig. 4a). When testing responses to first passes on consecutive days we found no difference (%RHR: $t_5 = -0.832$, $p = 0.443$; recovery time: $t_6 = -0.305$, $p = 0.770$; Fig. 4b). After nest-check experiments recovery time was significantly lower the second day (recovery time: $t_5 = 9.473$, $p = 0.001$), although no difference in heart rate reaction was found (%RHR: $t_5 = 0.693$, $p = 0.151$; Fig. 4c). For two birds we made nest-check recordings during three consecutive days, and found that recovery times were not further reduced but remained level (i.e. at 10 or 19 min) after the second experiment.

4. Discussion

Humboldt penguin breeding success varied with levels of human activity on the three islands of the Humboldt Penguin National Reserve. Penguins breeding on remote Chanaral had the highest breeding success, whereas accessible Choros showed reduced breeding success, and tourist-prone Damas had negligible reproductive output. These differences were significant. In comparison, Magellanic penguins breeding in the tourist sector at Punta Tombo, Argentina show no reduced breeding success, despite frequent visitation of nests at close proximity on occasion even touch of attending birds (Yorio and Boersma, 1992).

For situations other than nest desertion during significant disturbance, it is difficult to relate the observed breeding success directly to human activity. Many other factors such as weather, food availability, natural predation or parasites can influence the dynamics of individual populations. The three islands of the Humboldt Penguin National Reserve lie in close proximity but differ in size and habitat, thus it is important to consider factors other than disturbance which might have affected reproductive output.

4.1. Possible factors affecting breeding success

High abundance of external parasites can cause breeding failure in Peruvian seabirds (Duffy, 1983). However, external parasites were equally sparse at all sites and thus not responsible for differences in breeding success observed between islands.

Nor was natural predation of offspring an important factor on all three islands. Breeding failure due to human disturbance is often caused by evicting incubating birds from their nests exposing eggs or chicks to predators that would normally be driven off by the attending bird (e.g. Hockey and Hallinan, 1981; St. Clair and St. Clair, 1992). However, on Damas and Choros low abundance of gulls and raptors and the absence of other potential predators allowed eggs to be found untouched even months after nest abandonment. On Chañaral abundant Kelp Gulls *Larus dominicanus* might have inflated the observed breeding success as possible failures early in the breeding season would not have been detectable. Furthermore, Kelp Gulls may affect future breeding success on Chañaral if penguins are driven off nests.

Heavy rainfall can reduce Humboldt penguin breeding success (Simeone et al., 2002). However, the three islands lie in close proximity thus climatic conditions can be considered similar. Nest habitat and other nest attributes had no significant effect on breeding success.

Smaller colonies are sometimes thought to be new colonies with a high proportion of young breeders (e.g. Teneza, 1971). Although the age structure at different colonies is not known, anecdotal evidence suggests Damas' colony was already established and considerably larger some 15 years ago (Navarro, G., personal communication). Hence, Damas cannot be considered a new colony, and age and experience of breeding penguins are likely to be similar to Choros and Chañaral. Lack of experience should thus not be responsible for breeding failure on Damas.

During the guard phase Humboldt penguins forage up to 48 km from colonies, although they prefer to make shorter

trips when food is abundant (Hennicke, 2001). As Chañaral is less than 30 km north of the other two islands, foraging grounds theoretically overlap. Egg size, an indicator of food availability (e.g. Frere et al., 1998), was identical at all three islands, suggesting similar feeding conditions.

4.2. Effect of human disturbance on breeding success and nesting densities

Human disturbance is the most likely explanation for breeding failure of Humboldt penguins on Damas. All nests close to tracks failed, whereas successful nests were out of human sight. The 2002/03 season was an exceptionally successful breeding year at all other study areas, however, most nests on Damas failed, with only one chick surviving at the least accessible nest site. The fact that birds breeding at Damas never managed to raise two chicks may indicate human presence not only stressed the attending bird but could have prevented the foraging partner from returning and feeding the chicks.

On Damas all active nests were under rock boulders, thus people could not collapse nest burrows as was the main problem on the Puñihuil islands (South-central Chile (41° S), Simeone and Schlatter, 1998). Also overheating of eggs should not have been a problem even if penguins were driven off their clutch. Therefore, we suggest that cumulative stress caused by frequent visits possibly together with delayed return of partners was the main cause of nest desertion on Damas.

On Choros only a few active nests were found along the south-western plateau of the island despite abundant nesting opportunities: Of 710 nest sites found only 6% were occupied by breeders. In contrast, the rarely visited northern breeding area had an occupation rate of available nest sites (20%) similar to that found on Chañaral. Anecdotal evidence suggests the south-western plateau was used by more penguins in the 1990s (Marta Angarita, personal communication). This area is now often frequented by fishermen when traversing to the exposed western beaches of Choros, and by researchers heading for study sites.

Penguins breeding along the south-western plateau of Choros appeared to be calmer than average; together with Damas the south-western plateau was the only area where attending penguins never fled during nest-checks. Lower nest occupation rates together with lower and less variable behavioural and heart rate responses in penguins breeding along the south-western plateau suggest stress-prone birds may no longer attempt to breed in areas frequented by humans.

During the first study season, breeding success of experimental nests was higher than average, whereas during the second season it was lower. This can be explained by the high plasticity of breeding onset in Humboldt penguins: In the first breeding season we could subjectively select the "best" nest sites (sheltered one entrance nests with well-established birds) for disturbance experiments and thus observed good breeding success despite regular human presence. During the second season penguins started breeding earlier so we had to include three late breeders (all failed) to complete our experiments. Breeding success was unrelated to individual stress response, indicating that carefully regulated human presence (one approach per day over a short period of time) at

selected nests may not necessarily provoke breeding failure in Humboldt penguins. As has been shown for Adélie penguins (Giese, 1996), frequency of visitation, rather than type of intrusion, may be the critical factor influencing breeding success.

4.3. Disturbance experiments

Behavioural measures, such as the alternate stare, were correlated with physiological reaction, but insufficient to quantify stress in this study. Furthermore, behavioural measures can be misleading as penguins may show behavioural habituation to human presence despite an unaltered physiological stress response, as has been shown for Magellanic penguin chicks close to fledging (Walker et al., 2005). Heart rate telemetry allows evaluation of distinct disturbance stimuli in an unobtrusive and unbiased manner. Walking past a nest provoked significantly lower responses than did direct approaches. Angle of approach has been shown to be an important factor for disturbance tolerance in other species, and direct approaches are generally perceived more dangerous (Burger and Gochfield, 1990, 1991; Martin et al., 2004; but see Fernández-Juricic et al., 2005). Passing at 100 m distance or more caused less response than passing closer to the nest site. However, even passing at 150 m distance from an incubating penguin provoked a significant physiological response. These are the greatest response distances reported for penguins to date. Our findings support Sivak et al. (1987) in attesting that Humboldt penguins appear to have normal visual optics on land; they are definitely not short sighted as stated by Martin and Young (1984). A pass at 20 m distance from the nest but out of sight of the bird provoked similar heart rate responses as did passes at distances ≥ 100 m, indicating visual clues are more important for Humboldt penguins while acoustic stimuli of human foot steps on gravel alone are perceived less dangerous.

The Humboldt penguin is the most timid penguin species so far studied. Human approach caused heart rate acceleration in Adélie (Culik et al., 1990; Wilson et al., 1991), Gentoo (Nimon et al., 1995), Magellanic (Ecks, 1996), and African penguins (Marianne de Villiers, personal communication). Maximum heart rate measured during human approach to 2–3 m was 132 bpm for Gentoo and Magellanic penguins and 140 bpm for Adélie penguins or in percent of previously measured resting heart rate: 182%RHR for Magellanic, 157% for Gentoo and 184% for Adélie penguins. Humboldt penguins showed almost twice the response: maximum recorded heart rate was 258 bpm (287%RHR) during human approach to within 2 m from the nest (i.e. nest-check experiment). This is comparable to the stress response of Adélie penguins during capture and handling (270%RHR; Culik et al., 1990).

For proper evaluation heart rate responses to human disturbance must be related to natural species-specific responses. The great range of resting heart rates measured in Humboldt penguins (50–142 bpm) can be expected in a desert climate with temporally and locally variable ambient temperatures. Butler and Woakes (1984), who measured heart rate via implanted transmitters, reported mean heart rates of Humboldt penguins standing motionless on land (121 bpm), floating inactively on water (139 bpm), and during voluntary

dives (134 bpm); the highest values were recorded when the birds ran (245 bpm). This highly energy consuming behaviour provokes comparable heart rates to those observed in incubating penguins when approached to within 2 m. Natural disturbance stimuli during incubation usually provoked little deviation from RHR. The maximum recorded natural reaction was during partner return (174 bpm), and comparable to that caused by a human passing at 50 m distance.

However, more important than HR increase alone is the time the bird needs until the HR drops back to normal levels. During natural stimuli HR usually fell back to RHR within seconds; after human disturbance birds needed more time for recovery. Whereas in Royal penguins *Eudyptes schlegeli* HR returned to pre-approach values within 3 min (Holmes et al., 2005) and in African penguins as soon as the person retreated out of sight (Marianne de Villiers, personal communication), Humboldt penguins needed up to half an hour for recovery after experimental human approach. As heart rate in penguins can be directly correlated to energy consumption (e.g. Green et al., 2001), the long recovery phase in Humboldt penguins is presumably associated with high energetic costs.

Low level human disturbances (e.g. passes at over 100 m distance) are likely to accumulate, leading to negative physiological consequences such as increased energy demands. Thus, even if the birds are not forced to leave the nest immediately, repeated human visitation might increase the likelihood of nest desertion.

4.4. Habituation potential

Habituation can be defined as a “reduced response to repeated stimulation not attributable to fatigue or sensory adaptation” (Domjan, 2003). Although habituation to human disturbance has often been reported in colonial waterbirds, what is actually described is variation in tolerance between different colonies (Nisbet, 2000). Higher tolerance of human proximity in frequently disturbed colonies suggests that habituation of individuals has occurred. For example, African penguins showed reduced responses following regular disturbance on landing beaches (van Heezik and Seddon, 1990), and Magellanic penguins at Punta Tombo behaved differently depending on their previous exposure to people (e.g. Yorio and Boersma, 1992). However, recorded differences in tolerance might have been the consequence of shyer individuals leaving the area or failing to reproduce, as has been suggested repeatedly (e.g. Fowler, 1999, but see Walker et al., 2006; Fernández-Juricic, 2000; Müllner et al., 2004; this study).

Habituation to repeated human approach has been shown for Frigatebirds *Fregata minor*, Blue-footed Boobies *Sula nebouxii*, Swallow-tailed Gulls *Creagus furcatus* (Jungius and Hirsch, 1979), and Magellanic penguins (Walker et al., 2006). Habituation potential varies between species: Black ducks *Anas rubripes* showed less behavioural reaction after repeated exposure to aircraft noise, whereas wood ducks *Aix sponsa* did not habituate (Conomy et al., 1998).

Humboldt penguins appear to have little habituation potential to human disturbance. Each of three repeated passes at 20 m distance provoked similar heart rate responses and recovery times, as did experimental passes during consecutive days. However, recovery times after nest-check experiments

were significantly reduced after the same experiment one day later, even though the heart rate maximum did not change. This might suggest even Humboldt penguins are able to habituate to human activity to a certain degree, especially when stress response is costly (as is the case during long recovery phases after nest-check experiments) and birds do not receive any negative enforcement. Nevertheless, stress responses in Humboldt penguins remain extremely high in comparison with other species.

Magellanic penguins show reduced physiological and behavioural responses to repeated short and consistent human activity (Walker et al., 2006); whereas, three years of exposure to 1 h of unregulated visitation per day was insufficient to result in habituation (Fowler, 1999). Promoting habituation appears to require a maximum of predictable “low level disturbance”. For Humboldt penguins this could be passes at over 100 m distance from the nest site along established tracks or acoustic stimuli from well-regulated tourists behind visual barriers. However, Humboldt penguins are clearly unsuitable for tourist operations that include frequent, exposed human visitation at nest sites, as is the practice for several other penguin species including the closely related Magellanic penguin.

4.5. Survival of the shyest?

Seabirds, including Humboldt penguins, have been hunted by coastal human communities in Chile and Peru for more than 11,000 years (Sandweiss et al., 1998; DeFrance et al., 2001; Simeone and Navarro, 2002; DeFrance, 2005). Decimated populations of Humboldt penguins still are hunted in some areas (BirdLife International, 2005). Relative to their population size Magellanic penguins breeding along the sparsely populated southern and eastern coasts of South America never experienced selection pressure of similar magnitude. This might explain the higher tolerance of human disturbance observed in Magellanic penguins (Yorio and Boersma, 1992; Fowler, 1999; Walker et al., 2006).

Anti-predator behaviour has been shown to have heritable components (e.g. Roeder and Treat, 1961; Riechert and Hedrick, 1990; Boissy, 1995), and hunting pressure is likely to select for shyer individuals (McDougall et al., 2006). Conceivably, harvest of Humboldt penguins constituted selection for birds that are cautious of human approach. This might be an adequate strategy for a long-lived seabird when human disturbance is intermittent and irregular. However, with increasing pressure from tourism, these extreme responses become an evolutionary dilemma. Suppression of escape behaviour to minimize costs of disturbance when exposed to visitors could have severe consequences in other situations and may be counter-selected for (Müllner et al., 2004). Furthermore, extreme stress responses observed in Humboldt penguins suggest more tolerant individuals have been lost from the wild population.

4.6. Management implications

Properly managed tourism can be compatible with penguin breeding as has been shown for Magellanic (Yorio and Boersma, 1992) and Yellow-eyed penguins (Ratz et al., 1999).

However, marked inter-specific differences in responses to human disturbance mean generic seabird tourism management guidelines cannot be applied.

Good vision together with extraordinary shyness makes the Humboldt penguin a difficult focal species for ecotourism. During visual contact, visitors have to be guided clearly past and not approach penguins, as direct approaches were shown to provoke significantly stronger stress responses than that to tangential movements. Visual barriers are needed at distances of more than 100 m from breeding colonies to allow for human visitation without visual contact. As Humboldt penguins are very suspicious of any changes in their environment (returning birds may alter their path to avoid closer contact to any unknown structure at distances of more than 100 m, personal observation) visual barriers have to be elaborate to blend in with the environment. During moult any additional energy expenditure can promote starvation, thus even greater set-back distances (>150 m) should apply. Fortunately, Humboldt penguins do respond less to acoustical stimuli. A person passing at 20 m distance from the nest but out of sight of the bird provoked similar heart rate responses as did passes at 100 m and more. However, our experiments were performed by one single person in a calm steady manner, whereas tourism operations involve groups of noisy people, thus it is likely the distances at which disturbance occurs will be greater. Beale and Monaghan (2004) found that effects of disturbance increased with numbers of visitors, and decreased with distance from nests of two colonial seabird species. Thus, when managing access to wildlife areas, larger parties of visitors have to be kept further away or set-back distances need to be determined for the largest party likely to visit the site.

4.7. Conservation concern

Chañaral is the single most important breeding location of the Humboldt penguin (Mattern et al., 2004), and seems likely to face increasing human activity in the near future. The nearby fishing village Caleta Chañaral currently invests heavily in tourism, while the absence of any administrative bodies in the proximity of Chañaral could make the island vulnerable to unregulated human intrusion. The nearest ranger station of the Chilean Forestry Corporation (Corporación Nacional Forestal – CONAF, the agency responsible for administration of the islands) is located at distant Punta Choros. This distance, along with other constraints, makes adequate management of the reserve an ongoing challenge (Vilina et al., 1995; Mattern et al., 2004).

Acknowledgements

Financial support was provided by FONDECYT 1010250 (to GLJ), Stipendienstiftung University of Kiel, Germany, Deutsche Umwelthilfe and Global Nature Fund (to UE). Vaude and Enders kindly made expedition tents and aluminium boxes available. We thank Felipe Sepúlveda, Guido Navarro, Helmo Perez, Ingo Hahn, Jana Kotzerka, Katta Ludynia, Lars-Gunnar Ellenberg, Macarena Contreras, Marta Angarita, Mauro Cifuentes, Pato Ortiz, Pablo Tello, Pancho Díaz, Roberto

Villablanca, Stefan Günther and Tamara Ponce for assistance during fieldwork. José Miguel Torres (CONAF, IV Region) and Marcelo Zepeda (SERNAPECSA – Servicio Nacional de Pesca) provided in depths information about the Humboldt Penguin National Reserve. Access and experimental procedures were approved by CONAF IV Region and Subsecretaria de Pesca, Chile. Lars-Gunnar Ellenberg computed heart rate analysis software. Hermann Ellenberg, Alejandro Simeone, John Darby, and two anonymous reviewers provided valuable comments on drafts of this manuscript. We thank the organizations “arenaPH” and “CODEFF” in Chile for successfully campaigning against the exclusion of parts of the reserve for commercial exploitation.

REFERENCES

- Beale, C.M., Monaghan, P., 2004. Human disturbance: people as predation-free predators? *J. Appl. Ecol.* 41, 335–343.
- BirdLife International, 2005. Species factsheet: *Spheniscus humboldti*. Available from: <http://www.birdlife.org> Downloaded 18 September 2005.
- Boersma, P.D., Stokes, D.L., Yorio, P.M., 1990. Reproductive variability and historical change of magellanic penguins (*Spheniscus magellanicus*) at Punta Tombo, Argentina. In: Davis, L.S., Darby, J.T. (Eds.), *Penguin Biology*. Academic Press Inc., New York.
- Boissy, A., 1995. Fear and fearfulness in animals. *Q. Rev. Biol.* 70, 165–191.
- Boo, E., 1990. *Ecotourism: Potentials and Pitfalls*. WWF, Washington, DC.
- Boyle, S.A., Samson, F.B., 1985. Effects of non-consumptive recreation on wildlife: a review. *Wildl. Soc. Bull.* 13, 110–116.
- Burger, J., Gochfield, M., 1990. Risk discrimination of direct versus tangential approach by basking black iguanas (*Ctenosaura similis*): variation as a function of human exposure. *J. Comp. Psychol.* 104, 388–394.
- Burger, J., Gochfield, M., 1991. Human distance and birds: tolerance and response distances of resident and migrant species in India. *Environ. Conserv.* 18, 158–165.
- Butler, P.J., Woakes, A.J., 1984. Heart rate and aerobic metabolism in Humboldt penguins, *Spheniscus humboldti*, during voluntary dives. *J. Exp. Biol.* 108, 419–428.
- Carney, K.M., Sydeman, W.J., 1999. A review of human disturbance effects on nesting colonial waterbirds. *Waterbirds* 22, 68–79.
- Conomy, J.T., Dubovsky, J.A., Collazo, J.A., Flemming, W.J., 1998. Do black ducks and wood ducks habituate to aircraft disturbance? *J. Wildl. Manage.* 62, 1135–1142.
- Croxall, J.P., McInnes, S.J., Prince, P.A., 1984. The status and conservation of the seabirds at the Falkland Islands. In: Croxall, J.P., Evans, P.G.H., Schreiber, R.W. (Eds.), *Status and Conservation of the World's Seabirds*. International Council for Bird Preservation, Cambridge, UK.
- Culik, B.M., Wilson, R.P., 1991. Penguins crowded out? *Nature* 351, 340.
- Culik, B.M., Adelung, D., Woakes, A.J., 1990. The effects of disturbance on the heart rate and behaviour of Adélie penguins (*Pygoscelis adeliae*) during the breeding season. *Antarctic Ecosystems Ecological Change and Conservation*. Springer Verlag, Berlin. pp. 177–182.
- DeFrance, S., 2005. Late Pleistocene marine birds from southern Peru: distinguishing human capture from El Niño-induced windfall. *J. Arch. S.* 32, 1131–1146.
- DeFrance, S., Keefer, D., Richardson, J., Umire Alvaraez, A., 2001. Late Paleo-Indian coastal foragers: Specialized extractive behavior at Quebrada Tacahuay, Peru. *Latin American Antiquity* 12, 413–426.
- Domjan, M., 2003. *The Principles of Learning and Behaviour*, fifth ed. Wadsworth/Thomson Learning, Belmont, CA.
- Duffy, D.C., 1983. The ecology of tick parasitism on densely nesting Peruvian seabirds. *Ecology* 64, 110–119.
- Ecks, M., 1996. Einfluß von Störungen auf die Herzschlagrate brütender Magellanpinguine *Spheniscus magellanicus*. Friedrich-Wilhelms-Universität, Bonn.
- Edington, J.M., Edington, M.A., 1986. *Ecology, Recreation and Tourism*. Cambridge University Press, Cambridge.
- Eggleton, P., Siegfried, W.R., 1979. Displays of the Jackass Penguin. *Ostrich* 50, 139–167.
- Fernandéz-Juricic, E., 2000. Local and regional effects of pedestrians on forest birds in a fragmented landscape. *Condor* 102, 247–255.
- Fernandéz-Juricic, E., Venier, P.V., Renison, D., Blumenstein, D.T., 2005. Sensitivity of wildlife to spacial pattern or recreationist behaviour: A critical assessment of minimum approaching distances and buffer areas for grassland birds. *Biol. Conserv.* 125, 225–235.
- Fowler, G.S., 1999. Behavioral and hormonal responses of magellanic penguins (*Spheniscus magellanicus*) to tourism and nest site visitation. *Biol. Conserv.* 90, 143–149.
- Frere, E., Gandini, P., Boersma, P.D., 1998. The breeding ecology of magellanic penguins at Cabo Virgenes, Argentina: What factors determine reproductive success? *Colon. Waterbirds* 21, 205–210.
- Frid, A., Dill, L.M., 2002. Human caused disturbance stimuli as form of predation risk. *Conserv. Ecol.* 6, [online] URL: <http://www.consecol.org/vol6/iss1/art11>.
- Giese, M., 1996. Effects of human activity on Adélie penguin *Pygoscelis adeliae* breeding success. *Biol. Conserv.* 75, 157–164.
- Giese, M., Handworth, R., Stephenson, R., 1999. Measuring resting heart rates in penguins using an artificial egg. *J. Field Ornithol.* 70, 49–54.
- Green, J.A., Butler, P.J., Woakes, A.J., Boyd, I.L., Holder, R.L., 2001. Heart rate and rate of oxygen consumption of exercising macaroni penguins. *J. Exp. Biol.* 204, 673–684.
- Hennicke, J., 2001. Variabilität im Jagdverhalten des Humboldtpinguins (*Spheniscus humboldti*) unter verschiedenen ozeanographischen Bedingungen. In: *Mathematisch-Naturwissenschaftliche Fakultät*. Christian-Albrechts-Universität zu Kiel, Kiel, p. 104.
- Hockey, P.A.R., Hallinan, J., 1981. Effect of human disturbance on the breeding behaviour of Jackass penguins *Spheniscus demersus*. *S. Afr. J. Wildl. Res.* 11, 59–62.
- Holmes, N., Giese, M., Kriwoken, L.K., 2005. Testing the minimum approach distance guidelines for incubating Royal penguins *Eudyptes schlegeli*. *Biol. Conserv.* 126, 339–350.
- Honey, M., 1999. *Ecotourism and Sustainable Development*. Island Press, Washington, DC.
- Hüppop, O., 1995. Störungsbewertung anhand physiologischer Parameter. *Der ornith. Beob.* 92, 257–268.
- Ingold, P., Huber, B., Mainini, B., Marbacher, H., Neuhaus, P., Rawyler, A., Roth, M., Schnidrig, R., Zeller, R., 1992. Freizeitaktivitäten: Ein gravierendes Problem für Tiere? *Ornithol. Beob.* 89 (4), 205–216.
- IUCN (World Conservation Union), 2004. 2004 IUCN Red List of Threatened Species. Available from: www.redlist.org. Downloaded on 18 September 2005.
- Jungius, H., Hirsch, U., 1979. Herzfrequenzänderungen bei Brutvögeln in Galapagos als Folge von Störungen durch Besucher. *J. Ornithol.* 120, 299–310.
- Keller, V., 1995. Auswirkungen menschlicher Störungen auf Vögel – eine Literaturübersicht. *Ornithol. Beob.* 92, 3–38.
- Lee, A.K., McDonald, I.R., 1985. Stress and population regulation in small mammals. *Oxf. Rev. Reprod. Biol.* 7, 261–304.

- Levine, S., 2005. Developmental determinants of sensitivity and resistance to stress. *Psychoneuroendocrinology* 30, 939–946.
- Liddle, M.J., Scorgie, H.R., 1980. The effect of recreation on freshwater plants and animals: a review. *Biol. Conserv.* 17, 183–206.
- Lindberg, E., Hawkins, D.E., 1993. *Ecotourism: A Guide for Planners and Managers*. The Ecotourism Society, North Bennington, VT.
- Luna Jorquera, G., Garthe, S., Sepúlveda, F., Weichler, T., Waeskes, J.A., 2000. Population size of Humboldt penguins assessed by combined terrestrial and at sea counts. *Waterbirds* 23, 506–510.
- Martin, G.R., Young, S.R., 1984. The eye of the Humboldt penguin, *Spheniscus humboldti*: visual fields and schematic optics. *Proc. R. Soc. Lond., Ser. B: Biol. Sci.* 223, 197–222.
- Martin, J., de Neve, L., Fargallo, J.A., Polo, V., Soler, M., 2004. Factors affecting the escape behaviour of juvenile Chinstrap penguins, *Pygoscelis antarctica*, in response to human disturbance. *Polar Biol.* 27, 775–781.
- Mattern, T., Ellenberg, U., Luna Jorquera, G., Davis, L.S., 2004. Humboldt penguin census on Isla Chanaral, Chile: Recent increase or past underestimate of penguin numbers? *Waterbirds* 27, 368–376.
- McClung, M.R., 2004. Nature-based tourism impacts on Yellow-eyed penguins *Megadyptes antipodes*: does unregulated visitor access affect fledging weight and juvenile survival? *Biol. Conserv.* 119, 279–285.
- McDougall, P.T., Réale, D., Sol, D., Reader, S.M., 2006. Wildlife conservation and animal temperament: causes and consequences of evolutionary change for captive, reintroduced and wild populations. *Animal Conserv.* 9, 39–48.
- McEwan, B., 2000. Stress, definition and concepts of. In: Fink, G. (Ed.), *Encyclopaedia of Stress*, vol. 3. Academic Press, San Diego, pp. 508–509.
- Müllner, A., Linsenmair, K.E., Wikelski, M., 2004. Exposure to ecotourism reduces survival and affects stress response in hoatzin chicks (*Opisthocomus hoazin*). *Biol. Conserv.* 118, 549–558.
- Neebe, B., Hüppop, O., 1994. Der Einfluß von Störreizen auf die Herzschräge brütender Küstenseeschwalben (*Sterna paradisea*). *Artenschutzreport* 4, 8–13.
- Nimon, A.J., Stonehouse, B., 1995. Penguin responses to humans in Antarctica: some issues and problems in determining disturbance caused by visitors. In: Dann, P., Norman, I., Reilly, P. (Eds.), *The Penguins: Ecology and Management*. Surrey Beatty & Sons Pty Limited, Chipping Norton, NSW, Australia, pp. 420–439.
- Nimon, A.J., Schroter, R.C., Stonehouse, B., 1995. Heart rate of disturbed penguins. *Nature* 374, 415.
- Nimon, A.J., Schroter, R.C., Oxenham, R.C., 1996. Artificial eggs: measuring heart rate and effects of disturbance in nesting penguins. *Physiol. Behav.* 60, 1019–1022.
- Nisbet, I.C.T., 2000. Disturbance, habituation, and management of waterbird colonies. *Waterbirds* 23, 312–332.
- Ratz, H., Thompson, C., Brossy, J.J., Cooper, J., Crawford, R., Klages, N., Shannon, L., 1999. Who is watching whom? Checks for impacts of tourists on Yellow-eyed penguins *Megadyptes antipodes*. *Marine Ornithol.* 27, 205–210.
- Regel, J., Pütz, K., 1997. Effect of human disturbance on body temperature and energy expenditure in penguins. *Polar Biol.* 18, 246–253.
- Riechert, S.E., Hedrick, A.V., 1990. Levels of predation and genetically based anti-predator behaviour in the spider, *Agelenopsis aperta*. *Anim. Behav.* 40, 679–687.
- Roberts, C.L., Roberts, S.L., 1973. Survival rate of Yellow-eyed penguin eggs and chicks on the Otago peninsula. *Notornis* 20, 1–5.
- Roeder, K.D., Treat, A.E., 1961. The detection and evasion of bats by moths. *American Scientist* 49, 135–148.
- Ryel, R., Grasse, T., 1998. Marketing ecotourism: Attracting the elusive ecotourist. In: Whelan, T. (Ed.), *Nature Tourism: Managing for the Environment*. Island Press, Washington, DC.
- Sandweiss, D., McInnis, H., Burger, R., Cano, A., Ojeda, B., Paredes, R., Sandweiss, M., Glascock, M., 1998. Quebrada Jaguay: Early South American maritime adaptations. *Science* 281, 1830–1832.
- Simeone, A., Navarro, X., 2002. Human exploitation of seabirds in coastal southern Chile during the mid-Holocene. *Rev. Chil. Hist. Nat.* 75, 423–431.
- Simeone, A., Schlatter, R.P., 1998. Threats to a mixed species colony of *Spheniscus* penguins in southern Chile. *Colon. Waterbirds* 21, 418–421.
- Simeone, A., Araya, B., Bernal, M., Diebold, E.N., Grzybowski, K., Michaels, M., Teare, J.A., Wallace, R.S., Willis, M.J., 2002. Oceanographic and climatic factors influencing breeding and colony attendance patterns of Humboldt penguins *Spheniscus humboldti* in Central Chile. *Mar. Ecol. Prog. Ser.* 227, 43–50.
- Simeone, A., Luna Jorquera, G., Bernal, M., Garthe, S., Sepúlveda, F., Villablanca, R., Ellenberg, U., Contreras, M., Munoz, J., Ponce, T., 2003. Breeding distribution and abundance of seabirds on islands off north-central Chile. *Rev. Chil. Hist. Nat.* 76, 323–333.
- Sivak, J.G., Howland, H.C., McGill-Harestad, P., 1987. Vision of the Humboldt penguin (*Spheniscus humboldti*) in air and water. *Proc. R. Soc. Lond., Ser. B: Biol. Sci.* 229, 467–472.
- St. Clair, C.C., St. Clair, R.C., 1992. Weka predation on eggs and chicks of Fiordland crested penguins. *Notornis* 39, 60–63.
- Teneza, R., 1971. Behaviour and nesting success relative to nest location in Adélie penguins (*Pygoscelis adeliae*). *Condor* 73, 81–92.
- van Heezik, Y., Seddon, P.J., 1990. Effect of human disturbance on beach groups of jackass penguins. *S. Afr. J. Wildl. Res.* 20, 89–93.
- Vilina, Y.A., Capella, J.J., Gonzáles, J., Gibbons, J.E., 1995. Apuntes para la conservación de las aves de la Reserva Nacional de Pingüino de Humboldt. *Boletín Chileno de Ornitología* 2, 2–6.
- Walker, B.G., Boersma, P.D., Wingfield, J.C., 2005. Physiological and behavioral differences in magellanic penguin chicks in undisturbed and tourist-visited locations of a colony. *Conserv. Biol.* 19, 1571–1577.
- Walker, B.G., Boersma, P.D., Wingfield, J.C., 2006. Habituation of adult magellanic penguins to human visitation as expressed through behavior and corticosterone secretion. *Conserv. Biol.* 20, 146–154.
- Williams, T.D., 1995. *The Penguins, Spheniscidae*. Oxford University Press, Oxford.
- Wilson, R.P., Culik, B., Danfeld, R., Adelung, D., 1991. People in Antarctica – how much do Adélie penguins (*Pygoscelis adeliae*) care? *Polar Biol.* 11, 363–371.
- Woehler, E.J., Penney, R.L., Creet, S.M., Burton, H.R., 1994. Impacts of human visitors on breeding success and long-term population trends in Adélie penguins at Casey, Antarctica. *Polar Biol.* 14, 269–274.
- Yorio, P., Boersma, P.D., 1992. The effects of human disturbance on Magellanic penguin (*Spheniscus magellanicus*) behaviour and breeding success. *Bird Conserv. Int.* 2, 161–173.
- Zar, J.H., 1999. *Biostatistical Analysis*, 4th ed. Prentice Hall, New Jersey.
- Zavalaga, C.B., Paredes, R., 1997. Humboldt penguins at Punta San Juan, Perú. *Penguin Conserv.* 10, 6–8.